

Hanyong Xu

email hanyongx@mit.edu
website hanyongxu.com
address 77 Massachusetts Ave.
Building 9
Cambridge, MA 02139

RESEARCH INTEREST

- Responsible AI in urban science systems
- Platform economy, urban forms, and human behavior

EDUCATION

- 2023 - present **Ph.D., Urban Science** Cambridge, MA
Massachusetts Institute of Technology
- Advisor: Jinhua Zhao
 - *Selected Coursework: Optimization Methods; Transportation Foundation & Methods; Demand Modeling*
- 2019 - 2020 **Master, Urban Spatial Analysis** Philadelphia, PA
University of Pennsylvania
- Advisor: Ken Steif
 - *Selected Coursework: Spatial Statistics and Data Analysis; Data Wrangling and Visualization*
- 2015 - 2019 **Honors Bachelor of Arts** Toronto, ON
University of Toronto
- with High Distinction
 - Double Major in Economics and Architectural Design

WORKING PAPERS

- [1] **Xu, H.**, Zhao, J. “Longitudinal Evaluations of the Coverage and Pricing Differences between Transportation Network Companies and Traditional Taxis: A Case Study of New York City.”
- [2] He, X., **Xu, H.**, Shen C., Zhuang D., Zheng Y., & Zhao J. “Modeling Latent Demand and Reducing Prediction Disparities of Ride-hailing: A Fair Quantile Regression Method.”
- [3] Guo X., **Xu H.**, Zheng Y., Zhuang D., & Zhao J. “[Disparity-Reducing Vehicle Rebalancing in the Ride-hailing System.](#)”

CONFERENCES & WORKSHOPS

- Jan 2026 **Xu, H.**, Zhao, J. “Longitudinal Evaluations of the Coverage and Pricing Differences between Transportation Network Companies and Traditional Taxis: A Case Study of New York City.”
106th Transportation Research Board Annual Meeting (TRB, poster presentation, scheduled)

He, X., **Xu, H.**, Shen C., Zhuang D., Zheng Y., & Zhao J. “Modeling Latent Demand and Reducing Prediction Disparities of Ride-hailing: A Fair Quantile Regression Method.” *106th Transportation Research Board Annual Meeting (TRB, lectern presentation, scheduled)*

July 2025 Gao J., **Xu H.**, Dao L. “[Simulating Multi-Stakeholder Decision-Making with Generative Agents in Urban Planning](#)” *32nd International Conference on Transdisciplinary Engineering (TE2025, oral presentation)*

Jan./Apr. 2025 Zhuang D., **Xu H.**, Guo X., Zheng Y., Wang S., & Zhao J. “[Mitigating Spatial Disparity in Urban Prediction Using Residual-Aware Spatiotemporal Graph Neural Networks: A Chicago Case Study](#).” *105th Transportation Research Board Annual Meeting (TRB, poster presentation), The International Workshop on Spatio-Temporal Data Mining from the Web (WebsT’25 @ WWW’25, workshop, Best Paper Award) [proceeding]*

Oct. 2024 **Xu H.** “Navigating Algorithmic Unfairness in Ride-Hailing: Examining Disparate Impacts of Transportation Network Company Algorithms in New York City.” *2024 INFORMS Annual Meeting (invited talk)*

Sept. 2024 Mo B., **Xu H.**, Cho J. H., Zhuang D., Ma R., Guo X., & Zhao J. “[Large Language Model for Travel Behavior Prediction](#).” [\[extended abstract\]](#) *Conference in Emerging Technologies in Transportation Systems (TRC-30, poster presentation)*

RESEARCH EXPERIENCE

2023 - present	Researcher, JTL Urban Mobility Lab, MIT	Cambridge, MA
2022	Research Assistant, FUSE Lab, Hong Kong University	Remote

PROFESSIONAL EXPERIENCE

2021 - 2023	Data Analyst, Internal Risk Control and Compliance, Meituan	Beijing, China
2020 - 2021	Data and GIS Analyst, CityDNA Technology Co.	Beijing, China
2020	Data Science Intern, AreaProbe	Remote

TEACHING EXPERIENCE

2024 - 2025	Teaching Assistant, Introduction to Spatial Analysis and GIS, MIT
2024 - 2025	Teaching Assistant, Workshop on GIS, MIT

HONORS & GRANT

2024	Racially Just Research Initiative Microgrant, DUSP MIT
2024	Design and Technology Fellow , FASPE
2023	Presidential Graduate Fellowship , MIT
2020	Descartes Award (top 2 in the cohort) , University of Pennsylvania
2019	1st Place , Wharton Customer Analytics + Electronic Arts Datathon
2019	2nd Prize , Computational Design and Robotic Fabrication International Competition, DigitalFUTURES, Tongji University
2019	UofT Women in House , University of Toronto

INSTITUTE SERVICE

2024 - present **Student Representative**, MIT DUSP PhD Committee

SKILLS

Data Science & Machine Learning: Python + PyTorch, Julia + Gurobi, SQL, R, Excel, Google Cloud Computing

Front End & Visualization: JavaScript + html + css, Vue, Leaflet, Mapbox, Kepler

Geo-Spatial Analysis: ArcGIS, ArcPy, QGIS, Google Earth Engine, GeoDa

Product Design: Figma, Adobe Photoshop, Illustrator, InDesign

Last updated: Nov. 19th, 2025